Q1

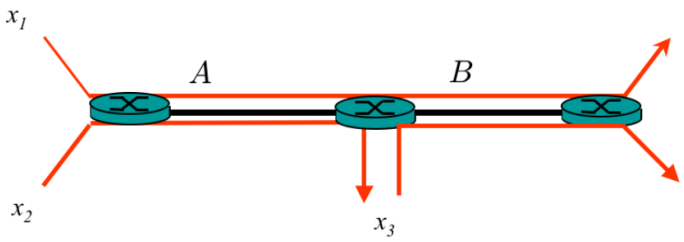


Figure 1: An example of network resource allocation over a two-link network with three flows. Link A has capacity 2 and link B has capacity 1.

of link B is 1. Suppose there are three users in the system x_1, x_2, and x_3 corresponding respectively to flows from node 1 to node 3, node 1 to node 2, and node 2 to node 3. There are several feasible

Under ppf what is the allocation

Q2

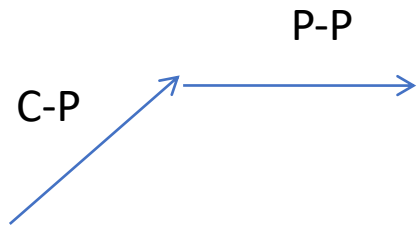
Let the utility function of user r be given by

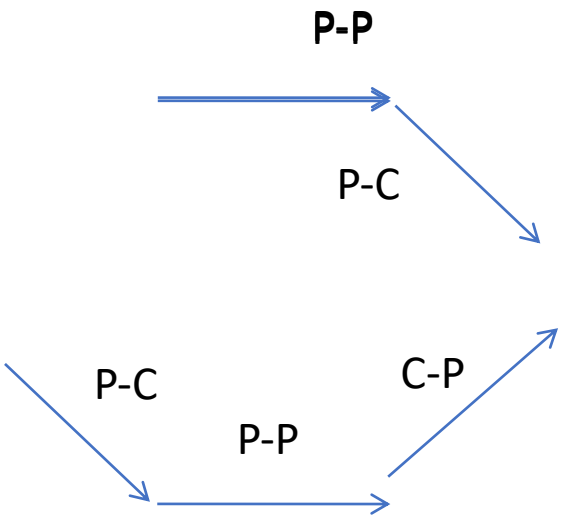
$$U_r(x_r) = w_r \frac{x^{1-\alpha}}{1-\alpha_r}$$

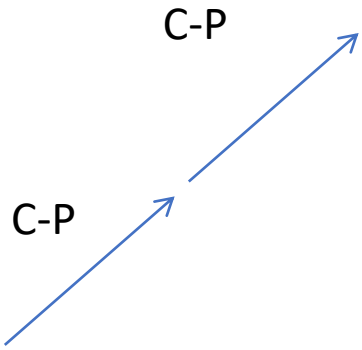
What is alpha for minimum **Potential delay fairness:**

Q3

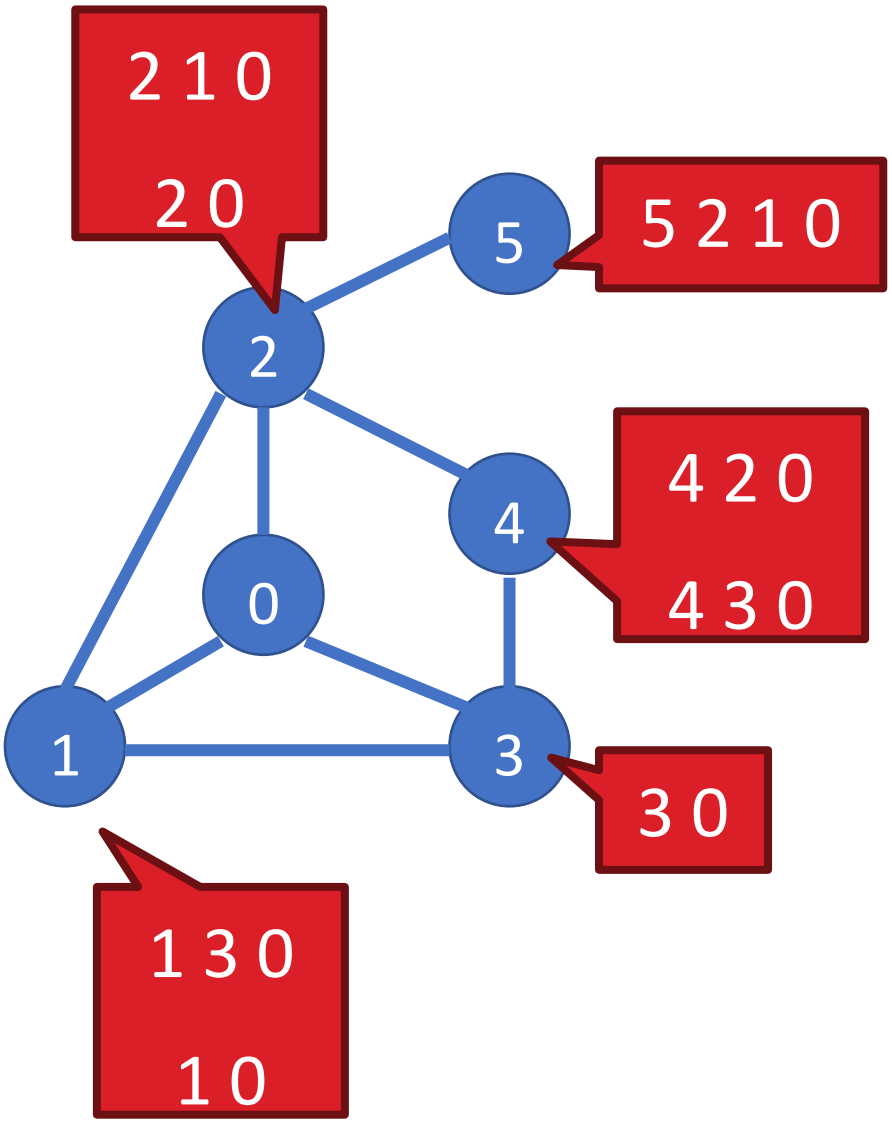
Which one is not valid







Q4: discuss about the following graphs? Stable , good gadget, bad gadget, naughty gadget, ..

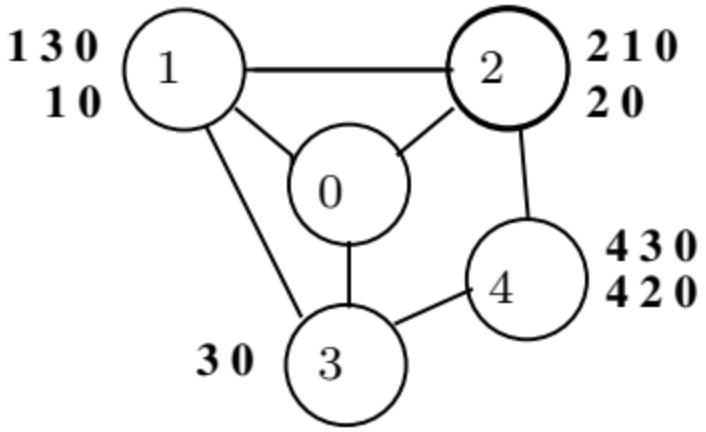


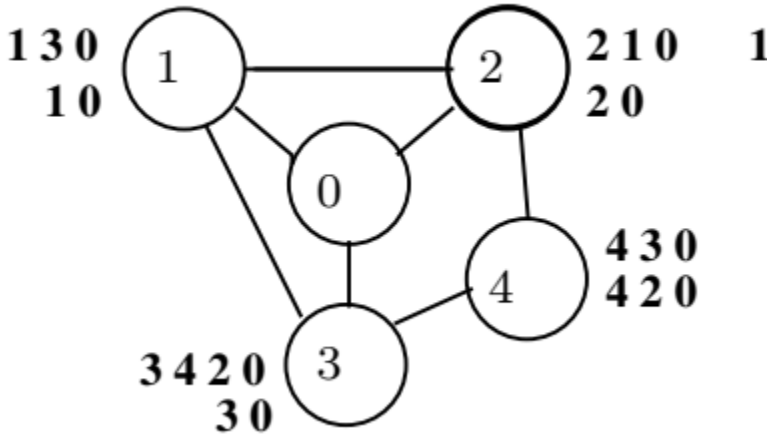
Graph of nodes and edges

Node 0, called the origin

A set of permitted paths from each node to the origin

Each set of paths is ranked


1 3 0
1 0
1
2
2 1 0
2 0
0
4 3 0
4 2 0
4
3 0
3


1 3 0
1 0
1
2
2 1 0
2 0
1
0
4 3 0
4 2 0
4
3 4 2 0
3 0
3